

BENTHIC MACROINVERTEBRATE RESPONSE TO HABITAT RESTORATION IN A CONSTRUCTED ARCTIC STREAM

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ABSTRACT

We examined the ability of engineered rock structures to enhance the productive capacity of benthic macroinvertebrates 2 years after their installation in a constructed stream in the Northwest Territories, Canada. We used a control-impact design to compare biological attributes of the structures to those of riffles in natural reference streams and among individual structures. The biomass of epilithon and coarse particulate organic matter, cover by bryophytes and macrophytes and volume of woody debris, however, were similar at structures and unenhanced control sites in the constructed stream and lower than at riffles in reference streams. Total macroinvertebrate density and densities of Chironomidae, Nematoda and Oligochaeta did not differ among engineered structures, unenhanced sites and reference riffles. In contrast, reference streams supported higher densities of Simuliidae and Ephemeroptera compared with engineered structures and unenhanced sites within the channel. Total biomass, biomass of Chironomidae, non-chironomid Diptera and combined biomasses of (i) Nematoda, Oligochaeta and Turbellaria and (ii) Ephemeroptera, Plecoptera and Hemiptera from reference riffles exceeded those at engineered structures. Density and biomass of macroinvertebrates varied within and among structure types and were highest at one of two ramps, one of two vanes and at the single groin compared with other structures. In the majority of cases, density and biomass of macroinvertebrates from individual structures were less than those at reference streams. These results suggest that engineered structures had variable effects on the productive capacity of benthic macroinvertebrates of the constructed streams and may be related to the extent that structures provided habitat conditions characteristic of reference riffles. © Her Majesty the Queen in Right of Canada 2011. Reproduced with the permission of the Minister of Natural Resources.

KEY WORDS: engineered structures; productive capacity; benthic macroinvertebrates; constructed stream; Arctic

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INTRODUCTION

Restoration ecologists use a variety of engineered structures to mitigate effects of habitat degradation or loss (Department of Fisheries and Oceans, 1986; Ebrahimnezhad and Harper, 1997; Laasonen *et al.*, 1998; Gerhard and Reich, 2000; Roni *et al.*, 2002). Habitat structures are typically designed to enhance fish production by providing a suitable combination of physical conditions that increase rates of food intake, provide velocity refugia or add cover (Minns, 1997; Jones and Tonn, 2004; Gabriel *et al.*, 2009). Surprisingly, few studies have evaluated the success of engineered structures to enhance fish production mediated through enhancement of their benthic macroinvertebrate prey (Reeves *et al.*, 1991; Bradshaw, 1996; Minns *et al.*, 1996; Harper *et al.*, 1998; Jones and Tonn, 2004).

In 1991, diamonds were discovered in the remote region of the Northwest Territories of Canada known as the Barrenlands. In preparation for mineral extraction, the

southern portion of Panda Lake and Koala Lake were drained, and as a part of the habitat compensation agreement, a 3.4-km constructed stream was created. Since 1997, water from the non-drained northern portion of Panda Lake has been diverted through this constructed stream to Kodiak Lake. This stream restored connectivity for fish migrations and provided spawning, rearing and nursery habitats for the predominant Arctic grayling (*Thymallus arcticus*) (Jones *et al.*, 2003b). Our assessments of the performance of the constructed stream during the first 3 years after its creation have shown that it was not a good surrogate for natural reference streams. Poor performance of the constructed stream likely results from continued inputs of mud, silt and sand from the steep, physically unstable and largely non-vegetated banks that limit accrual of allochthonous and autochthonous energy sources. Densities of benthic macroinvertebrates in the constructed stream were well below those in reference streams (Jones *et al.*, 2003a; Jones and Tonn, 2004; Jones *et al.*, 2008).

To enhance secondary production to levels comparable with reference streams, a suite of engineered structures were deployed in late 1998 to create riffle habitats. The extent to which these engineered structures enhance primary and secondary production shortly after their deployment is poorly

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known. Understanding how these recently deployed structures enhance the productive capacity of the constructed stream is important for calculating compensation ratios (Minns, 2006) to achieve the Department of Fisheries and Oceans' long-term policy objective of no-net-loss of productive capacity of fish habitats (Department of Fisheries and Oceans, 1986). Furthermore, evaluations of the effectiveness of engineered structures to enhance productive capacity of benthic macroinvertebrate communities in Arctic settings are absent. Their effectiveness in northern environs is expected to differ from their temperate counterparts because high latitude streams are nutrient poor and experience ice cover or complete freezing for long periods, which reduce macroinvertebrate growth rates and extend generations times (Cummins and Merritt, 1996; Wallace and Anderson, 1996; Jones *et al.*, 2003a; Jones *et al.*, 2010). In the present study, we report on the initial effectiveness of engineered structures to increase the productive capacity of the constructed stream 2 years after their installation. We used a control-impact design in which we compared the habitat characteristics and the density and biomass of benthic macroinvertebrates between (i) unaltered control habitats in the constructed stream, (ii) habitats modified by the engineered structures, and (iii) riffle habitats from reference streams. Lastly, we compared the habitat characteristics and the biological communities among the nine individual structures to assess their individual performance.

METHODS

Study area

The Ekati Diamond Mine is located about 310 km northeast of Yellowknife in the Southern Arctic ecozone of the Northwest Territories, Canada (Figure 1). The landscape is underlain by granite, gneiss and schists that form broad sloping uplands, plateaus and lowlands. Thin soils within the upland zones support American dwarf birch (*Betula glandulosa*) and scattered low shrubs, whereas lowland soils support sedge-tussock and low-tall shrub tundra underlain by continuous permafrost (Bliss, 1981). Large areas of low topography and extensive glaciations have moulded a landscape dominated by shallow lakes and connecting streams. The mean annual temperature is -12°C with relatively high summer (27°C) and low winter temperatures (-50°C) (Meteorological Service of Canada, 2004). Streams are typically frozen for 8 months of the year (i.e. October to May).

The constructed stream, engineered structures and reference streams

To allow mining of the diamond-bearing kimberlite ore, engineers established an ice dam so that the southern portion of Panda Lake and Koala Lake could be drained and mined.

After the establishment of the ice dam, a 3.4-km channel was created in 1997 to divert flow from the northern portion of Panda Lake to Kodiak Lake (Figure 1). The majority of the constructed stream is composed of a single channel (stream width = 4 to 6 m) constrained by steep ($10\text{--}90^{\circ}$) and high (2–10 m) banks. The constructed stream consisted of a 2.7-km segment with a gradient of 0.1% and a shorter 0.7-km segment with a gradient of 1.1%. Summer base flow typically ranges between 0.05 and $0.20\text{ m}^3\text{ s}^{-1}$ (Dillon, 2001).

In 2000, blast rock lined a largely non-vegetated riparian zone. Instream habitats in the stream were dominated by slow, deep, run and flat habitats, where fine sediments of silt, clay and sand comprised about 44% of the river bed (Jones, 2004). In contrast, fines on average covered only 14% of river bed materials of three reference streams (Polar-Vulture A, Polar-Vulture B, Pigeon), where riffle habitats characterized by cobbles and boulders predominate (Jones, 2004). Relative to reference streams, the constructed stream supported low biomass of aquatic bryophytes and macrophytes and low levels of habitat diversity (Jones *et al.*, 2008). As a result, managers concluded that efforts to improve the habitat conditions should consist of adding habitat structures that create faster-flowing riffle and run habitats.

Engineered rock structures, comprising vanes, weirs, ramps and a single groin (Roni *et al.*, 2005; Mooney *et al.*, 2007) were built in late 1998 with aggregate removed from eskers in the region. Because large woody debris is absent from reference streams in this region, no wood was used in structure construction. The design of each structure and their general influence on water velocity and depth are summarized in Table I and Figure 2. The primary objective of the structures was to convert slow-flowing run habitats to faster-flowing riffle habitats by constraining and modifying the cross-sectional area of the stream channel. In turn, the resulting increased water velocities would remove smaller-size fractions of silts and sands and foster the predominance of larger substratum size fractions of gravels and cobbles in the river bed (Figure 2).

Earlier analyses showed that the five reference streams are comparable with the constructed stream based on their physical and hydrologic characteristics (Jones *et al.*, 2008). The mean concentrations of the total phosphorus in the constructed stream ($9.9\text{ }\mu\text{g L}^{-1}$) are comparable with those in the natural streams ($7.0\text{ }\mu\text{g L}^{-1}$), whereas concentrations of the total nitrogen in the constructed stream ($321\text{ }\mu\text{g L}^{-1}$) are almost double that in the reference streams ($167\text{ }\mu\text{g L}^{-1}$) (Jones *et al.*, 2003b).

The constructed stream (length = 3.4 km) is, however, about 2.4 times longer than the average of the five reference streams (mean length $\pm 1\text{ SE} = 1.4 \pm 0.5\text{ km}$). Because the constructed stream was blasted out of granite, vegetation cover in the riparian zones in 2000 was minimal ($<5\%$ coverage) and dominated by large boulders and cobble. In

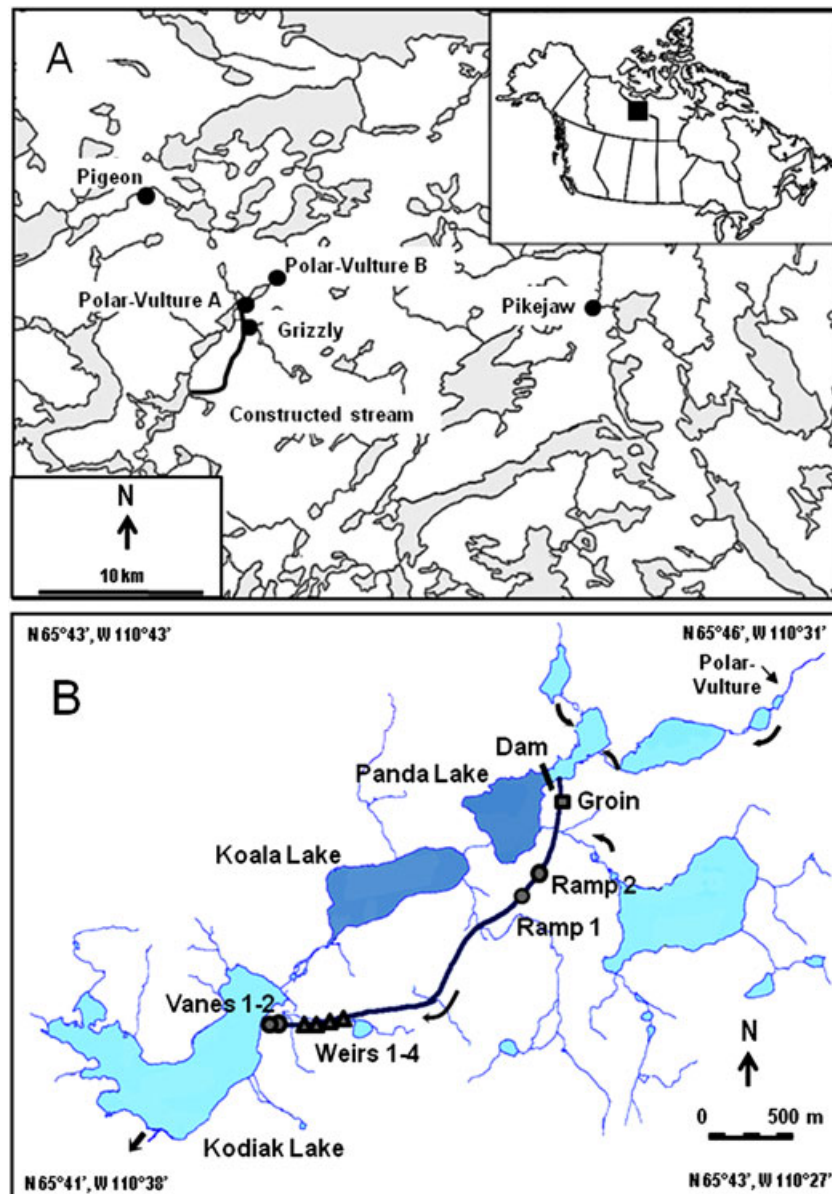


Figure 1. Location of the five natural streams (filled circles) used as reference sites, and the constructed stream (heavy line) (A), and location of the engineered structures within the constructed stream (B) in the Northwest Territories, Canada. Arrows indicate directions of flow within the Kodiak Lake drainage system. Unaltered control sites in the channel were located within 50 to 200 m of the structures. This figure is available in colour online at wileyonlinelibrary.com/journal/rra.

contrast, riparian zones of reference streams are well vegetated with American dwarf birch, willow (*Salix* spp.) and grasses covering >90% of riparian zones (G. Scrimgeour, University of Alberta, Edmonton, Alberta, unpublished results). Minimally vegetated riparian zones limit inputs of coarse and fine woody debris and organic matter to the constructed stream, whereas the steep and unstable banks consisting of large volumes of unconsolidated materials result in appreciable inputs of fine sediments to the constructed stream (G. Scrimgeour, unpublished data). Additional descriptions of

the five reference streams and the constructed stream are provided in Jones *et al.* (2003a, 2003b, 2008).

Study design

We used a control-impact design and a separate suite of comparisons among structures to assess their initial effectiveness (i.e. in 2000, 2 years after deployment) to enhance the productive capacity of the constructed stream. All

Table I. Habitat characteristics and hydraulic effects of engineered structures added to the constructed stream (see also Figure 2)

Structure	Description	Dominant habitat type(s) created by the structure	Influence on water velocity and depth and river bed materials
Ramps (2)	Raised stream bed comprising large boulders, cobbles and gravels that reduces depth and cross-sectional area of the stream	Riffle	Reduced water depth, increased water velocity, predominance of cobbles and gravels and areas of supercritical flow
Weirs (4)	Large boulders form an upstream pointing chevron that reduces cross-sectional area	Run-riffle and pool	Water flow is focused to the centre of the channel, creating turbulent water immediately downstream of the structure
Vanes (2)	Five series of large boulders create finger-like projections perpendicular to stream bank	Riffles interspersed with areas of shallow backwater pools	Deflection of the thalweg to the opposite side of the stream results in areas of reduced depth and increased water velocity interspersed with pockets of shallow backwater and pools with increased water depth and reduced velocity
Groin (1)	Single series of large boulders form a finger-like projection angled at about 45° downstream to flow	Riffle, pool, deeper backwater and slow run	Deflection of the thalweg to the opposite side of the stream results in medium velocity riffle towards the outside of the structure, and areas of increased depth that forms a pool-backwater and run immediately downstream of the structure

The numbers of each structure added to the stream are provided in brackets. Modified from Jones and Tonn (2004). Although structures modify downstream habitats, they have minimal effects on upstream habitat because of impounding.

comparisons were based on field collections taken in July (i.e. mid-summer).

Our control-impact comparisons contrasted habitat characteristics and biological communities at nine structures (two ramps, two vanes, four weirs, one groin) with those at nine unaltered control sites within the constructed stream (lower velocity flat and run habitats) and with riffle habitats in five nearby reference streams (Grizzly, Polar-Vulture A, Polar-Vulture B, Pigeon, Pikejaw; Figure 1). Unaltered sites in the constructed stream were located 50–200 m of the structures.

A potential concern of using a control-impact design is that differences between treatments (i.e. control sites versus impact sites) can reflect pre-existing conditions and not effects of the manipulation. Preliminary analyses, however, showed that mean water depths, velocity and proportions of fine (silts and sand) and coarse (cobble and gravels) substrate, measured in 1998 (prior to deployment of structures), did not differ between unaltered control sites and adjacent impact sites that subsequently received structures (*t*-tests, $p > 0.05$; G. Scrimgeour, University of Alberta, Edmonton, Alberta, unpublished results). Thus, any differences in physical, and by inference, biological, characteristics of control and impact sites identified in 2000 should reflect effects of the structures.

Abundance of benthic macroinvertebrates in lake outlet streams are often negatively related to distance downstream of the lake outlet, associated with declining levels of lake-derived seston (Richardson and Mackay, 1991; Harding, 1997). We controlled for these longitudinal patterns by ensuring that all sample sites in the constructed stream and in reference streams were located at least 100 m downstream of lake outlets.

We completed two additional sets of analyses to quantify variation in the performance of the nine structures. For these analyses, we compared (i) habitat characteristics and density and biomass of benthic macroinvertebrate communities among all nine structures and (ii) density and biomass of benthos supported by each of nine individual structures against those of the five reference streams.

Sampling methods

We described size fractions of river bed materials at each site (i.e. study reaches of 40 to 150 m within the constructed stream and five reference streams) by describing substrate composition at three to five equally spaced points along three to five transects perpendicular to streamflow. At each point along a transect, we visually classified the proportion of bed materials as clay and silt (<0.0625 mm), sand (0.0625–2 mm), small gravel (2–32 mm), large gravel (32–64 mm), cobble (64–256 mm) and boulder (>256 mm). We also recorded water depth (± 1 cm) and velocity (± 0.1 cm s⁻¹ at 0.6 times water depth) with a Marsh McBirney Flow mate (Model 2000) at four or five equally spaced points along each of the three to five transects. A surveyor's level and graduated rod were used to ensure that stream gradients based on water surface elevations in reference streams were similar to those at study sites in the constructed stream (Jones, 2004).

For control-impact comparisons, we determined the average substratum size fractions and water depth and velocity by averaging the 9 to 25 within-site measures. By contrast, within-site measures of river bed materials and water velocity and depth were retained for assessments of individual

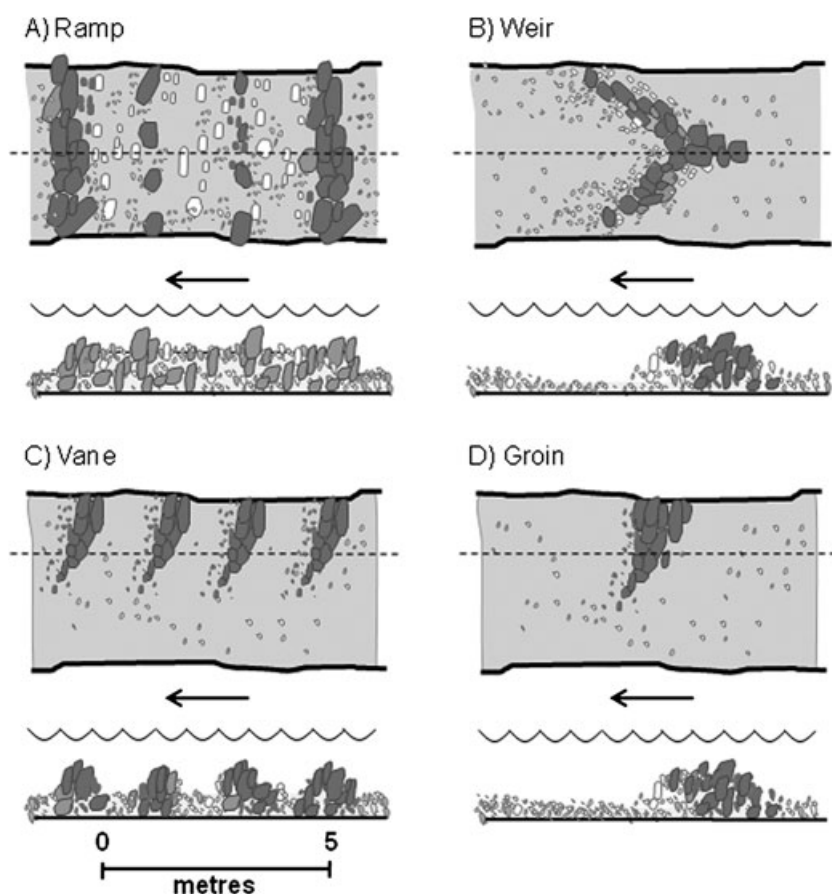


Figure 2. Generalized planar and side views of the engineered ramps (A), weirs (B), vanes (C) and groin (D) in the constructed stream. Arrows indicate general direction of water movement.

structures. We used the same approach for comparisons of coarse particulate organic matter (CPOM), epilithon, proportion coverage by bryophytes and macrophytes, and density and biomass of benthic macroinvertebrates (numbers of samples for specific comparisons are described below).

Four or five samples of CPOM were collected at each site by inserting an 8-cm-diameter plastic jar, 9-cm deep, vertically into the substratum. Within 48 h, we washed the refrigerated samples, removing inorganic material and invertebrate cases and exuviae, and sieved them through a 1-mm mesh screen. Filtered organics were dried to constant weight at 40 °C and weighed to the nearest 0.1 mg.

Four or five samples of epilithon were collected at each site by scraping the upper surfaces (4.9 or 9.8 cm²) of randomly chosen stones. We scraped smaller areas when algal biomasses were determined visually to be moderate or high and larger areas (created by pooling two smaller samples) when epilithic biomass was less abundant. Samples were frozen for 2–4 weeks in the dark before being dried, weighed, ashed at 550 °C and re-weighed to calculate ash-free dry mass. We quantified the abundance of bryophytes and macrophytes at each sampling site visually by estimating

coverage by these plants of the river bed within three to five 0.1-m² areas along each of three to five transects.

Reference streams support low volumes of wood composed predominantly of fine woody debris (maximum lengths < 1 m, maximum diameter < 10 cm) derived from dwarf birch and willow. Given its low abundance and small size, fine woody debris is a potential source of allochthonous matter, but not a strong modifier of benthic habitats in our study streams. Volumes of fine woody debris were determined by counting all pieces longer than 10 cm along all study reaches. We measured the length and diameter at each end of all pieces and calculated volumes, assuming that woody debris approximated a cylinder. The concentration of woody debris within a stream reach, standardized as cm³/100 m², was determined by dividing the total volume of woody debris in the stream reach by its surface area.

Benthic macroinvertebrates were collected using a 0.1-m² Surber sampler fitted with a 250-µm mesh net. Five samples were taken from reference streams and between three and five samples from each of the engineered structures and the unaltered sites in the constructed stream. All samples were preserved in 70% ethanol in the field. Invertebrates

were subsequently identified to genus or species, with the exception of Nematoda, Turbellaria and terrestrial invertebrates, which were typically identified to family or order. After enumeration, samples were dried to constant mass at 40 °C and weighed (± 0.1 mg) to estimate invertebrate dry mass. To ensure sufficient biomass for measurement, we combined several taxonomic groups (e.g. Nematoda, Oligochaeta, Turbellaria; Ephemeroptera, Plecoptera, Hemiptera; non-chironomid Diptera) into composite samples.

Statistical analyses

For our control-impact analyses, we used multiple samples taken within a site to create a single average datum for each site, that is, the nine unaltered control sites, the nine engineered structures and the five reference streams. Measures of water depth and velocity and proportion of fine and coarse substrate, macrophytes and bryophytes were taken from three to five points along each of the three to five transects at each site. We averaged the three to five measures along each transect and then created the single datum from the average of the transects. Measurements of epilithic biomass, biomass of CPOM and benthic macroinvertebrate density and biomass were based on averages of four or five within-site samples. Lastly, because we measured all woody debris at each site, a within-site average was not used.

To reduce type 1 error rates, we completed two one-way MANOVAs to test for significant multivariate main effects of treatment (i.e. unaltered control sites in the constructed, engineered structures in the constructed stream and reference streams) on the nine habitat descriptors (i.e. independent variables: water depth, velocity, proportion of fine and coarse substrate, epilithic biomass, CPOM, woody debris, proportions of bryophytes and macrophytes) and the 11 descriptors of benthic macroinvertebrate density and biomass (total density and densities of Chironomidae, Simuliidae, Nematoda, Oligochaeta and Ephemeroptera, total biomass and biomasses of Chironomidae, non-chironomid Diptera, Nematoda + Oligochaeta + Turbellaria, Ephemeroptera + Plecoptera + Hemiptera). We completed the two MANOVAs to consider habitat and benthic invertebrate effects separately and to ensure that the numbers of observations within individual cells approximated or exceeded the number of dependent variables. When a MANOVA was statistically significant (Wilks' lambda, $p < 0.05$), we tested for differences in habitat characteristics and densities and biomasses of macroinvertebrates among (i) unaltered control sites ($n=9$), (ii) sites containing engineered structures ($n=9$) and (iii) reference stream riffle habitats ($n=5$) using one-way ANOVAs. When the ANOVA was statistically significant ($p < 0.05$), we applied Bonferroni adjustments to p values for multiple comparisons using the least significant difference tests. For control-impact analyses,

we used a p value of 0.017 to adjust for inflated type 1 errors (0.05/3 comparisons).

We also completed two MANOVAs to test for differences in the six habitat characteristics (water depth, water velocity, epilithon, CPOM and proportions of fine and coarse substrate) and densities and biomasses of select taxonomic groups (total density and density of Chironomidae, Simuliidae, Nematoda, total biomass and biomass of Chironomidae, non-chironomid Diptera, Nematoda + Oligochaeta + Turbellaria) among the nine individual structures. For these analyses we used the multiple within-site samples as the unit of replication. Numbers of observations within each cell ranged from three to five. When a MANOVA was statistically significant (Wilks' lambda, $p < 0.05$), we tested for differences in habitat and macroinvertebrates among the nine structures using one-way ANOVAs, as mentioned previously. When the one-way ANOVA model was statistically significant ($p < 0.05$), we used a p value of 0.001 to adjust for inflated type 1 errors (0.05/45 comparisons).

We compared densities and biomasses of numerically dominant taxonomic groups from individual structures to those from reference riffles using one-sample t -tests. For these analyses, we compared replicated estimates of density and biomass from the five reference streams to the overall mean derived from the individual structures. We used a p value of 0.006 to adjust for inflated type 1 errors (0.05/9 comparisons for each group). For all statistical tests, we used the Kolmogorov–Smirnov test to examine data for normality and Levene's median test for homogeneity of variances and transformed data appropriately ($\log_{10}$ or arcsine square root) where required.

RESULTS

Comparisons with reference streams

Our surveys of the constructed stream showed that it consisted of a 2.7-km segment with a gradient of 0.1% and a shorter 0.7-km segment downstream with a gradient of 1.1%. Stream gradients in reference streams were similar (0.8–1.7%) to that in the constructed stream.

One-way MANOVA revealed a significant multivariate main effect of treatment (i.e. differences among unaltered sites, engineered structures and reference streams) on habitat descriptors (Wilks' lambda = 0.18, $F_{(9,13)} = 6.6$, $p = 0.001$). Individual ANOVA's showed that water depth, velocity and proportions of coarse substrate did not differ among the three treatments (Table II, Figure 3). By contrast, unaltered control sites and structure-enhanced sites contained significantly higher proportions of fine substrate (pairwise comparisons of non-altered controls and structures versus reference streams: $p = 0.001$ and 0.004; Bonferroni adjusted alpha: $p = 0.017$; Figure 3, Table II). Preliminary comparisons had shown that

Table II. Summary of one-way ANOVA tests evaluating the main effect of site type [i.e. unaltered channel sites ($n=9$), engineered structures ($n=9$) and riffles from reference streams ($n=5$)] on habitat and environmental attributes, and on density and biomass of benthic macroinvertebrates

Variable type	ANOVA	
	<i>F</i>	<i>p</i>
Habitat and environmental		
Water depth	2.6	0.102
Water velocity	2.9	0.070
Proportion silt and sand	9.4	0.001
Proportion gravel and cobble	2.2	0.135
Epilithon	88.9	0.001
CPOM	47.2	0.001
Woody debris	21.8	0.001
Proportion bryophytes	13.8	0.001
Proportion macrophytes	40.9	0.001
Benthic macroinvertebrate density and biomass		
Total density	1.6	0.234
Density of Chironomidae	1.3	0.297
Density of Simuliidae	6.2	0.001
Density of Nematoda	1.6	0.219
Density of Oligochaeta	1.7	0.226
Density of Ephemeroptera	21.8	0.001
Total biomass	14.3	0.001
Biomass of Chironomidae	5.4	0.013
Biomass of non-chironomid Diptera	11.0	0.001
Biomass of Nematoda, Oligochaeta and Turbellaria	8.3	0.003
Biomass of Ephemeroptera, Plecoptera and Hemiptera	37.1	0.001

Degrees of freedom for all models = 2, 20. Statistically significant main effects of site type ($p < 0.05$) are highlighted in bold. CPOM = coarse particulate organic matter.

mean water depth, velocity and proportion of fine and coarse substrate at sites where engineered structures were to be deployed did not differ from unaltered controls in the constructed stream (paired t -test for all comparisons, $df=18$, $p > 0.05$). Thus, subsequent differences in water depth, velocity and substratum size composition were likely due to the deployment of structures. Compared with reference riffles, stream beds from engineered structures and control sites supported lower biomasses of epilithon, CPOM and volumes of woody debris and were covered with less bryophytes and macrophytes (all pairwise comparisons of non-altered controls and structures versus reference streams: $p=0.001$; Bonferroni adjusted alpha: $p=0.017$) (Table II, Figure 3).

Total density and densities of Chironomidae, Nematoda and Oligochaeta from the engineered structures resembled those at both reference riffles and unaltered control sites in the constructed stream (Table II, Figure 4). Densities of Simuliidae and Ephemeroptera at engineered structures and at unaltered control sites, however, were lower than at reference riffles (pairwise comparisons of non-altered controls

and structures versus reference streams: $p=0.001$ to 0.005 ; Bonferroni adjusted alpha: $p=0.017$). ANOVA tests also revealed statistically significant differences in total biomass and biomass of all numerically dominant taxa or taxonomic groups among structures and reference streams (Table II). Total biomass and biomasses of individual taxonomic groups at structures were lower than in reference streams (ranges in all pairwise comparisons: $p=0.001$ to 0.016 ; Bonferroni adjusted alpha: $p=0.017$) (Figure 4). Biomass at unaltered control sites in the constructed stream was either similar to or greater than at structures (Figure 4).

Comparisons among engineered structures

Analyses of variance revealed statistically significant differences in water depth, velocity and size fractions of river bed materials but not epilithic biomass or CPOM (Table III, Figure 5). Mean water velocity was typically higher at ramp 2 and both vanes compared with other structures (Bonferroni adjusted alpha: $p < 0.001$). River bed materials at the four weirs and at the groin were dominated by silt and sand compared with the ramps and vanes, where substrates were dominated by gravel and cobble (Bonferroni adjusted alpha: $p < 0.001$; Figure 5). Woody debris was not observed at any of the nine structures.

We also found differences in mean density and biomass of benthic macroinvertebrates among individual structures (ANOVA, $p < 0.05$; Figure 5). Mean density and biomass were highest at one of the ramps, vanes and/or at the groin compared with other structures (Bonferroni adjusted alpha: $p < 0.001$; Figure 5). Mean density at the engineered groin exceeded all other structures (Bonferroni adjusted alpha: $p < 0.001$) and reflected high densities of Chironomidae and nematodes compared with other structures. Density of Chironomidae at the groin exceeded total densities at other structures. Density of Simuliidae at ramp 2 and the two vanes exceeded that at all other structures (Bonferroni adjusted alpha: $p < 0.001$) and coincided with the highest mean water velocity and lowest water depth. Mean density of Simuliidae did not differ among these three structures (Range of p values = 0.12 to 0.93), whereas density of Nematoda at the groin exceeded those at all other structures (Bonferroni adjusted alpha: $p < 0.001$). Statistically significant differences in densities of Oligochaeta or Ephemeroptera were not observed among structures (ANOVA, $p > 0.05$) (Figure 5).

Total faunal biomass and biomass of Chironomidae and non-chironomid Diptera, but not other taxonomic groups, also differed among structures (Table III, Figure 5). Total faunal biomass was typically highest at ramp 2, vane 1 and the groin and lowest at the four weirs (Bonferroni adjusted alpha: $p < 0.001$). Biomass of Chironomidae at the groin far exceeded that at all other structures, and biomass at ramp 1 exceeded those at all four weirs. Biomass of non-chironomid

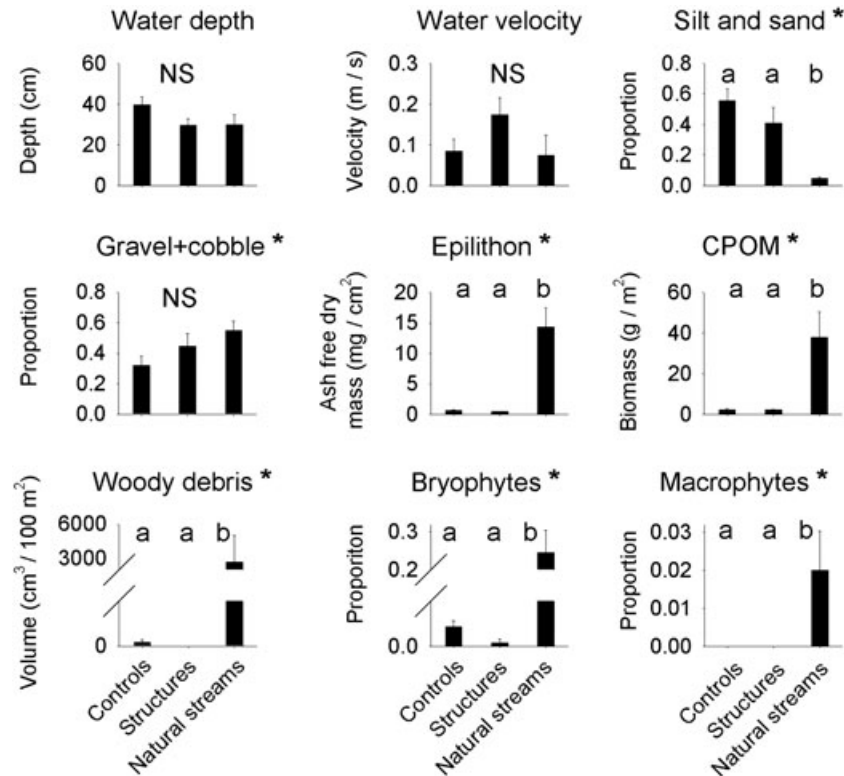


Figure 3. Mean (± 1 SE) water depth, water velocity, proportion of silt and sand, proportion gravel and cobble, biomass of epilithon and coarse particulate organic matter (CPOM), volume of coarse woody debris and proportion cover by bryophytes and by macrophytes from unaltered pool-run habitats in the constructed stream (controls), engineered structures, and from riffle habitats in natural streams. NS = statistically non-significant ($p > 0.05$); * = statistically significant ANOVA test ($p < 0.05$); histograms sharing the same letter are not significantly different based on Bonferroni adjusted p values.

Diptera was higher at ramp 2 and vane 1 compared with other structures (Figure 5).

Comparisons of individual structures with reference streams

One-sample t -tests showed that in the majority of cases (61 of 90 comparisons involving 10 measures of biomass and density for each of the nine comparisons of structures versus the reference streams), total density and biomass, and density and biomass of the dominant taxonomic groups at individual structures were lower than at reference streams (Bonferroni adjusted alphas: $p < 0.006$). Twenty-eight of the remaining 30 comparisons revealed non-significant differences in density and biomass between reference streams and individual structures (Table IV), including mean densities of oligochaetes at eight of the nine structures (but not weir 4) [$\log_{10}(x+1)$ transformed mean densities (± 1 SD) from reference streams = 1.76 ± 0.32 ; range of means from the eight individual structures = 1.4 to 2.0]. The one remaining comparison reflected the sole instance where total density at a structure (i.e. the sole groin) exceeded that in reference streams (Table IV).

DISCUSSION

Our control-impact analyses showed that despite low channel slopes, engineered structures increased local water velocities and the proportion of coarser river bed materials. Epilithic biomass, biomass of CPOM, volumes of woody debris and cover by macrophytes and bryophytes at the engineered by structures did not differ from unaltered control sites and were well below levels seen in reference streams. Engineered structures overall exerted highly variable effects on benthic macroinvertebrate communities. In the vast majority of cases, density and biomass of macroinvertebrates at structures were low relative to riffles in reference streams.

Lower densities and biomasses of most macroinvertebrate groups in the constructed stream (both altered and unaltered sites) likely reflects the paucity of food resources of epilithon, CPOM and fine woody debris for grazers and shredders, and indirectly for collector-gatherers. Epilithic mats in reference streams accrue appreciable biomass over the summer months supported by modest mean daily water temperatures (10°C), high water clarity (nephelometric turbidity

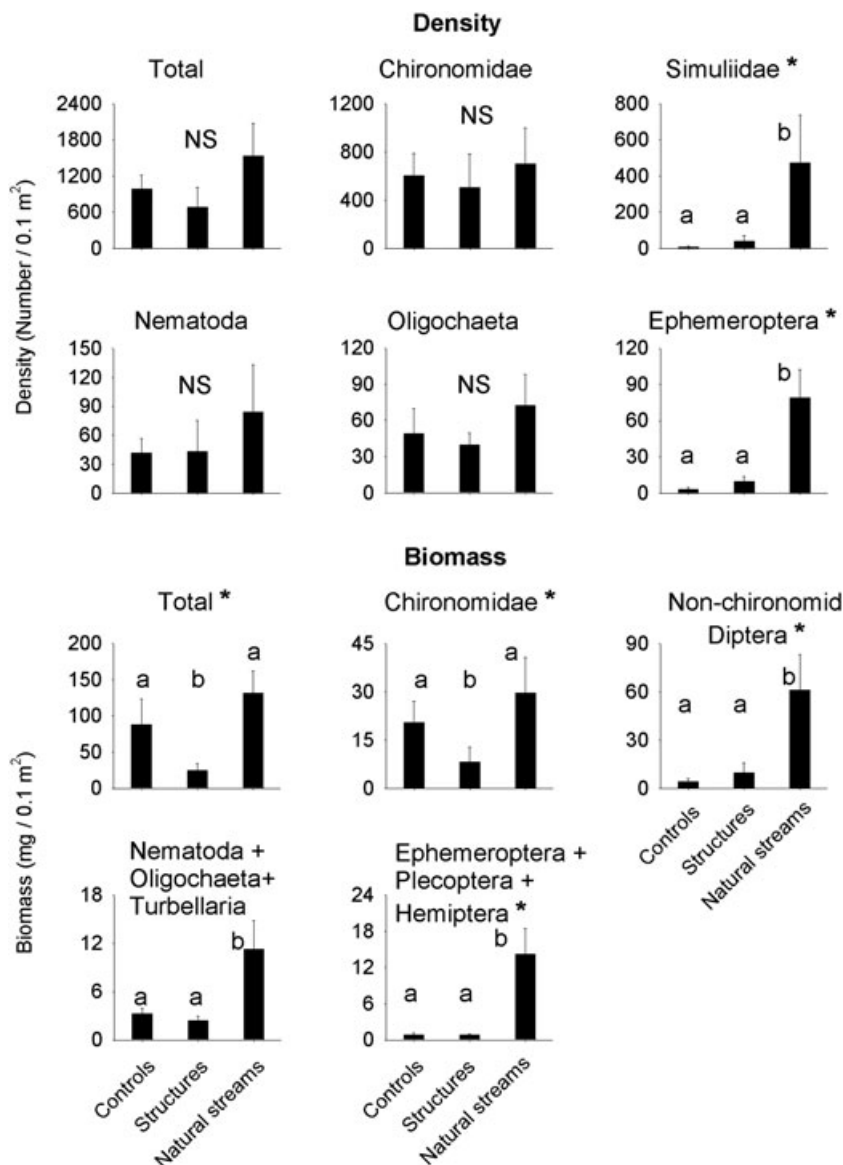


Figure 4. Mean ($\pm$ SE) total density (above) and biomass (below), and density and biomass of dominant macroinvertebrate taxa, from unaltered habitats in the constructed stream (controls), engineered structures, and riffle habitats in natural streams. NS = statistically non-significant ANOVA ($p > 0.05$); * = statistically significant ANOVA test ($p < 0.05$), histograms sharing the same letter are not significantly different based on Bonferroni adjusted p values.

levels < 0.5) and long day lengths (18 to 20 h day⁻¹) (Jones *et al.*, 2003a, 2003b). Although the constructed stream experienced similar environmental conditions over the summer, mean biomass of epilithon in the constructed stream was at least 30-fold lower than in reference streams. Low biomasses of epilithic mats in the constructed stream compared with reference stream are unlikely due to differences in nutrients because concentrations of total phosphorus and nitrogen in the constructed stream (phosphorus = 9.9 $\mu\text{g L}^{-1}$, nitrogen = 321 $\mu\text{g L}^{-1}$) are equivalent or higher than that in reference streams (total phosphorus = 7.0 $\mu\text{g L}^{-1}$, total nitrogen = 167 $\mu\text{g L}^{-1}$) (Jones

et al., 2003b). We expect that high loadings of fine sediments from the river banks and mobilization of these materials reduce algal accrual in the constructed stream. In addition, low inputs of CPOM and woody debris from poorly developed riparian vegetation likely explain low levels of CPOM and woody debris in the constructed stream.

A constraint of our evaluation is that our assessment was conducted about 2 years after deployment of the engineered structures. Given low rates of colonization and numerical responses of benthic macroinvertebrates in highly disturbed or recently formed habitats (Mackay, 1992), differences in

Table III. Summary of one-way ANOVA tests evaluating differences in habitat and environmental attributes, and in density and biomass of benthic macroinvertebrates among the nine engineered structures in the constructed stream

Variable type	ANOVA		
	<i>F</i>	<i>df</i>	<i>p</i>
Habitat and environmental variables			
Water velocity	23.9	8,31	0.001
Water depth	13.4	8,31	0.001
Proportion silt and sand	31.9	8,18	0.001
Proportion gravel and cobble	33.1	8,18	0.001
Epilithon	0.63	8,30	0.75
Coarse particulate organic matter	1.21	8,31	0.32
Benthic macroinvertebrate density and biomass			
Total density	14.5	8,27	0.001
Density of Chironomidae	29.7	8,27	0.001
Density of Simuliidae	10.1	8,27	0.001
Density of Nematoda	16.0	8,27	0.001
Total biomass	8.1	8,27	0.001
Biomass of Chironomidae	15.2	8,27	0.001
Biomass of non-chironomid Diptera	5.8	8,27	0.001
Biomass of Nematoda, Oligochaeta and Turbellaria^a	2.4	8,27	0.041

Analyses are based on three to five replicate measures at each of the nine structures. Statistically significant main effects of structure type ($p < 0.05$) are highlighted in bold.

^aIndicates statistically significant ANOVA model but statistically non-significant pairwise comparisons using Bonferroni adjusted alpha.

benthic communities between structures and reference streams were expected to exist. Despite this short period, density and biomass of larval Chironomidae and Simuliidae at some individual structures resembled those in reference riffles. Larval Chironomidae and Simuliidae are widely recognized as early colonizers after flow disturbances (Mackay, 1992) and are also among the first groups to become established in constructed streams. For instance, Malmqvist *et al.* (1991) documented the abundance of larval Simuliidae during the first 18 months after the establishment of a constructed stream in Sweden. They also reported initial high rates of colonization by filter feeders, and that selected characteristics of the constructed stream resembled those of reference outlet streams within 1 year.

Our structure-specific comparisons showed that ramp 2, vane 1 and the single rock groin were more effective at supporting select taxa than were the other six structures. High densities of larval filter-feeding Simuliidae at ramp 2 and both vanes were accompanied by the creation of shallow, high-velocity riffles that provide crucial conditions for filter feeders. River bed materials at all three structure were dominated by boulder and large cobbles, and the upper surfaces of these substrata supported high densities of filter feeders, in part because they provide stable attachment sites compared with the more mobile surrounding matrix of

gravels and sands, which were largely devoid of simuliids. Enhanced density and biomass of Simuliidae at ramp 2 could also reflect the close proximity of these particular structures to the lake outlet (e.g. Harding, 1992), which can provide high densities of lake-derived seston that are a prime food resource for Simuliidae (Richardson and Mackay, 1991). However, the two vanes that also supported high densities of Simuliidae are in the lower section of the constructed stream. This suggests that the constructed stream is capable of supporting high densities of filter-feeding Simuliidae though its entire length provided that suitable substrate and hydraulic conditions are present.

High density and biomass of larval Chironomidae at the rock groin, located within 100 m of the lake outlet, may have resulted from inputs of lake-derived seston, which created a high-quality food resource for collector-gathering Chironomidae. Although not quantified, we also observed high biomasses of epilithic and epipelic algae mats in the 100-m reach immediately upstream of the Groin to the lake outlet. It is possible that these communities also contributed organic matter to the groin and that stable flow regimes and the presence of slower, deeper water areas created by the groin favour retention of these materials.

The underlying reasons for the lower performance of ramp 1 and the four weirs remain unclear. We suspect that the low performance of ramp 1 arises because it did not constrain water flow sufficiently to increase water velocities. In fact, mean water velocity at ramp 1 (0.07 m s^{-1}) was slightly lower than that at unaltered sites (0.08 m s^{-1}) and was about five times lower than that at the highly productive ramp 2 (0.39 m s^{-1}). The inability of the four weirs to enhance benthic macroinvertebrate communities also coincided with low water velocities (range = $0.005 - 0.023 \text{ m s}^{-1}$) and river beds dominated by silts and sands (range = 50% to 64%), and conditions at the weirs were similar to that unaltered pool-run habitats.

Deployment of engineered structures was intended to increase productive capacity of the constructed stream by providing a greater diversity of habitats, including those appropriate for young-of-year (YOY) Arctic grayling. Our data suggest that engineered structures successfully altered physical conditions at some sites in the constructed stream by converting small areas of slow-moving, largely depositional habitat to faster-flowing erosional riffle habitats. Although converted habitats attracted significantly higher densities of YOY Arctic grayling (Jones and Tonn, 2004), mass and growth rates of fish in the constructed stream remained lower than in reference streams (Jones *et al.*, 2003b; Jones and Tonn, 2004). Our data suggest that reduced individual performance of fish in the constructed stream was most likely due to the inability of recently deployed structures to support enhanced densities and biomasses of their benthic macroinvertebrate prey. Whether or not the recently deployed

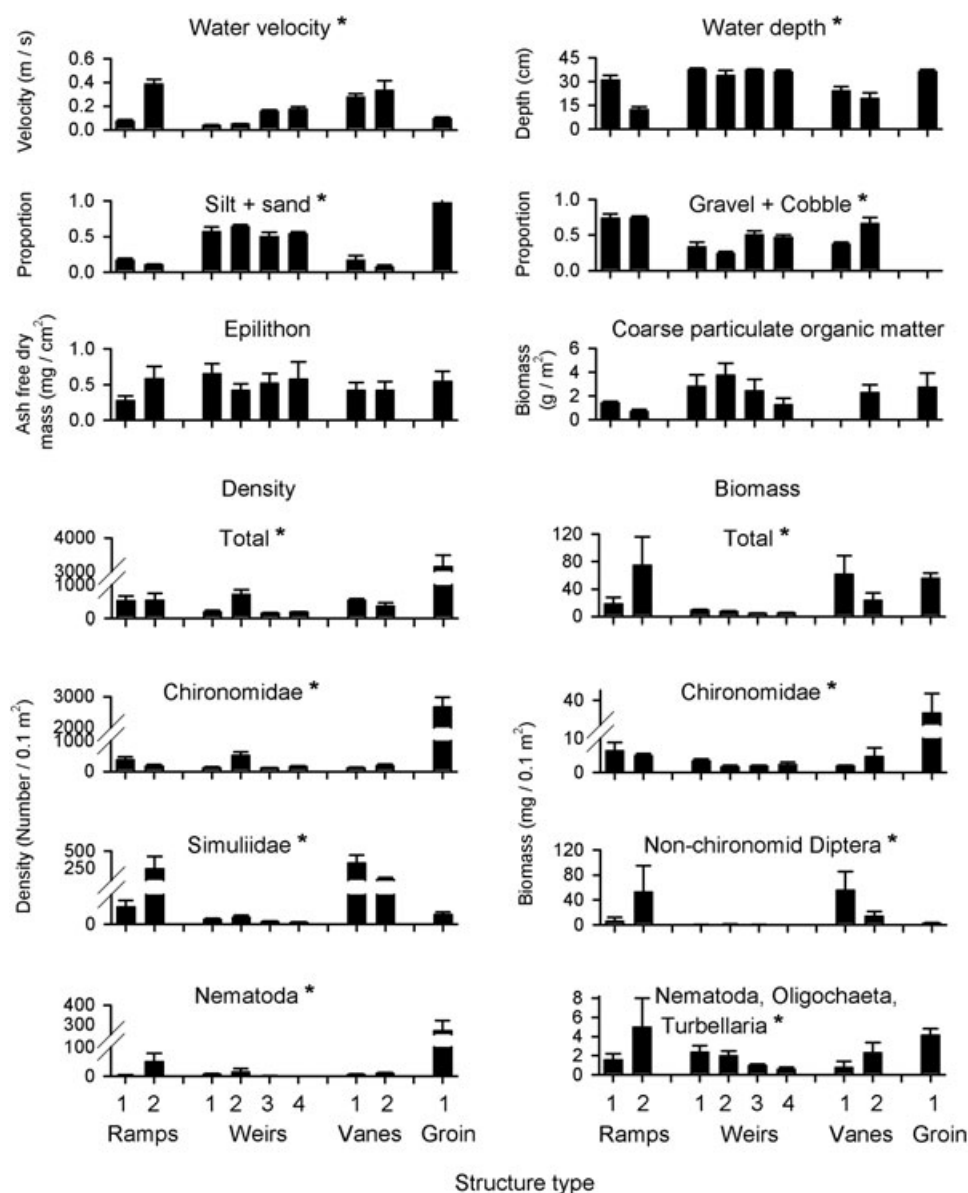


Figure 5. Mean (± SE) water velocity, water depth, proportions of silt and sand, proportions of gravel and cobble, total faunal density and biomass and density and biomass of predominant macroinvertebrate groups from individual engineered structures (vanes, weirs, ramps and a groin) within the constructed stream. * = statistically significant ANOVA test ($p < 0.05$).

structures could eventually support enhanced densities and biomasses of benthos requires additional monitoring.

Conservation biologists use a diversity of designs to evaluate the effectiveness of engineered structures on aquatic habitats and the biota that they support. We evaluated the effectiveness of engineered rock structures using control-impact comparisons (e.g. Lepori *et al.*, 2005) and comparisons of individual structures with reference streams. This combined approach allowed us to document the extent to which structures altered physical and hydraulic habitat and the structure of benthic communities and to gain insights on

variation in the performance of individual structures. The latter were important in understanding the inability of our control-impact comparisons to detect differences between (i) structures and unaltered control sites within the constructed stream and (ii) structures and to reference streams. Improvements in productive capacity due to the deployment of engineered structures may be subtle and/or highly variable at best during the first few years after deployment; furthermore, a broad suite of statistical comparisons may be required to detect and understand the effectiveness of engineered structures.

Table IV. Summary of one-sample *t*-tests comparing mean $\log_{10}$ or $\log_{10}(x+1)$ transformed density and biomass of benthic macroinvertebrates derived from five reference streams (± 1 SD) with mean density and biomass from individual structures

Comparison	Structure	Mean (± 1 SD) from reference riffles	Mean from structures	<i>t</i>	<i>p</i>	Conclusion
Total density	Ramp 1	3.09 $\pm$ 0.31	2.72	2.7	0.055	Ramp 1 = Reference
	Ramp 2	3.09 $\pm$ 0.31	2.73	2.6	0.06	Ramp 2 = Reference
	Weir 2	3.09 $\pm$ 0.31	2.85	1.7	0.16	Weir 2 = Reference
	Groin	3.09 $\pm$ 0.31	3.50	5.8	0.004	Groin > Reference
Density of Chironomidae	Ramp 1	2.71 $\pm$ 0.36	2.57	0.9	0.42	Ramp 1 = Reference
	Weir 2	2.71 $\pm$ 0.36	2.72	0.04	0.97	Weir 2 = Reference
	Vane 2	2.71 $\pm$ 0.36	2.27	2.8	0.051	Vane 2 = Reference
	Groin	2.71 $\pm$ 0.36	3.43	4.5	0.01	Groin = Reference
Density of Simuliidae	Ramp 2	2.45 $\pm$ 0.49	2.23	2.2	0.38	Ramp 2 = Reference
	Vane 1	2.45 $\pm$ 0.49	2.46	0.05	0.95	Vane 1 = Reference
	Vane 2	2.45 $\pm$ 0.49	1.84	2.8	0.051	Vane 2 = Reference
Density of Ephemeroptera	Ramp 2	1.83 $\pm$ 0.28	1.58	2.0	0.12	Ramp 2 = Reference
	Vane 1	1.83 $\pm$ 0.28	1.54	2.3	0.08	Vane 1 = Reference
Density of Nematoda	Ramp 2	1.70 $\pm$ 0.47	1.70	0.02	0.99	Ramp 2 = Reference
	Weir 2	1.70 $\pm$ 0.47	1.17	2.52	0.07	Weir 2 = Reference
Total biomass	Ramp 2	2.07 $\pm$ 0.22	1.87	2.04	0.11	Ramp 2 = Reference
Biomass of Chironomidae	Ramp 1	1.36 $\pm$ 0.35	1.14	1.4	0.24	Ramp 1 = Reference
	Groin	1.36 $\pm$ 0.35	1.57	1.38	0.24	Groin = Reference
Biomass of Non-chironomid Diptera	Ramp 2	1.66 $\pm$ 0.39	1.73	0.39	0.72	Ramp 2 = Reference
	Vane 1	1.66 $\pm$ 0.39	1.75	0.50	0.64	Vane 1 = Reference
Biomass of Nematoda + Oligochaeta + Turbellaria	Ramp 2	1.03 $\pm$ 0.26	0.78	2.19	0.09	Ramp 2 = Reference

Bonferroni adjusted alpha for pairwise comparisons = 0.006.

In contrast to woody debris (Grant *et al.*, 1986; Crispin *et al.*, 1993; Culp *et al.*, 1996), the effectiveness of rock structures to enhance stream habitats is poorly known (but see Negishi and Richardson, 2003; Lepori *et al.*, 2005), and their utility in Arctic settings is even less well understood (Muotka and Laasonen, 2002). In the absence of previous studies, resource managers at the Ekati Diamond Mine installed four structure types to create riffle habitats in the upper and lower sections of the constructed stream. Given the high construction costs and logistical challenges of installing structures during the short low-flow period, this approach resulted in low replication and poor spatial distributions of individual structure types. Despite these constraints, our data suggest that rock structures can sometimes create riffle habitat and support benthic communities similar to those from reference streams.

Management applications

The extent that local-scale enhancements can increase the overall productive capacity of the entire constructed stream depends on the extent to which they create favourable habitat and the amount of habitat that they comprise in the constructed stream. Individual engineered structures ranged from 20 to 40 m in length and accounted for a small portion (<10%) of the entire 3.4-km constructed stream. Under these conditions, it is unlikely that structures could balance

the losses of productive capacity of reference streams due to dewatering activities. The extent that deployment of additional or more effective engineered structures could increase productive capacity of the entire constructed stream warrants additional review and assessment.

The ability of the engineered structures to enhance the productivity of benthic macroinvertebrates likely resides in their capacity to (i) retain fine particulate organic matter and CPOM, (ii) provide stable river bed materials that can support epilithon as a food source and (iii) create areas of higher water velocity so support filter feeders, primarily larval simuliids, to utilize instream and lake-derived seston. The effectiveness of individual structures to enhance productivity through these routes may currently be compromised because larger channel-scale processes result in high levels of sediment inputs and sediment transport within the constructed stream, low amounts of inputs and retention of woody debris and CPOM. High inputs of sediments and mobile substrates in the constructed channel also likely explain low coverage of macrophytes. The effectiveness of the engineered structures could be improved by stabilizing the river banks by installing synthetic fabrics, accompanied by an intensive riparian planting programme. Planting of sedge-tussock and low-tall shrubs (including willow) would also provide a source of CPOM to the constructed stream that would enhance the detrital-based component of the stream food web.

Although our analyses provided some suggestions, our study was not specifically designed to explain variance in the performance of the different structure types. To improve our ability to achieve stream habitat restoration goals, we need additional studies to better understand the effectiveness of the structures. Such studies need to include detailed measurements of physical habitat and hydraulic attributes that structures create and establish linkages between invertebrate communities and the development of allochthonous and autochthonous energy sources. Ultimately, this information should allow resource managers and regulators to better understand the extent to which such structures can contribute to habitat compensation goals.

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